

ESP32 Free Wi-Fi Captive Portal + Video

Complete Project Rebuild, Setup, Troubleshooting & Reference Guide

Platform: ESP32 **IDE:** Arduino IDE 2.3.10 **Board:** ESP32 Dev Module

NOTE: This guide records the project setup and the problems encountered during the build, so it can be rebuilt after a long gap.

1. Project Overview

This project turns an ESP32 into a Wi-Fi Access Point. A phone connects to the ESP32 Wi-Fi, the ESP32 provides a local captive-portal webpage, and the webpage serves an MP4 video stored in the ESP32 LittleFS filesystem. Internet access is not required between the phone and ESP32.

```
Phone
↓
ESP32 Wi-Fi
↓
DNS captive portal
↓
Local webpage: 192.168.4.1
↓
/video.mp4 request
↓
LittleFS
↓
Video sent to phone
```

The original project archive contains FreeWiFi.ino and data/video.mp4. It also contains macOS metadata such as __MACOSX and .DS_Store, which are not needed.

2. Hardware Required

- ESP32 development board
- USB data cable
- Computer/laptop
- Smartphone for testing
- MP4 video file

No external sensors or additional electronic components are required.

3. Software Required

- Arduino IDE 2.3.10 (version used during setup)
- ESP32 board package by Espressif Systems
- ESPAsyncWebServer by ESP32Async
- AsyncTCP by ESP32Async
- Arduino LittleFS Upload extension

The COM number can change. In the original setup the board appeared as COM7.

4. ESP32 Board Package

Arduino IDE → Tools → Board → Boards Manager → search **esp32** → install **ESP32 by Espressif Systems**. Then select Tools → Board → ESP32 Arduino → ESP32 Dev Module.

5. Required Libraries

```
#include <WiFi.h>
#include <DNSServer.h>
#include <ESPAsyncWebServer.h>
#include <LittleFS.h>
```

- ESPAsyncWebServer by ESP32Async
- AsyncTCP by ESP32Async

NOTE: For ESP32 use AsyncTCP. Do NOT use ESPAsyncTCP for this project. ESPAsyncTCP is associated with ESP8266 projects.

6. Project Folder Structure

```
FreeWiFi/  
■■■ FreeWiFi.ino  
■■■ data/  
■■■ video.mp4
```

NOTE: The data folder must be beside the .ino file. LittleFS Upload uses this folder as the filesystem source.

7. Error: ESPAsyncWebServer.h Missing

```
fatal error: ESPAsyncWebServer.h: No such file or directory
```

Cause: Arduino IDE could not find ESPAsyncWebServer. Fix: install ESPAsyncWebServer by ESP32Async and AsyncTCP by ESP32Async.

8. Error: MbedTLS MD5 Functions

```
mbedtls_md5_starts_ret was not declared  
mbedtls_md5_update_ret was not declared  
mbedtls_md5_finish_ret was not declared
```

These errors appeared in ESPAsyncWebServer/src/WebAuthentication.cpp and indicate a compatibility problem between an installed async web-server library and the ESP32/MbedTLS environment.

The fix used was to remove the problematic async library versions and install the ESP32Async versions of ESPAsyncWebServer and AsyncTCP.

9. Removing Old Libraries with Windows CMD

```
cd /d "C:\Users\<YOUR_USERNAME>\Documents\Arduino\libraries"  
rmdir /s /q "ESPAsyncWebServer"  
rmdir /s /q "ESPAsyncTCP"  
rmdir /s /q "AsyncTCP"
```

NOTE: If a folder does not exist, CMD may say it cannot find it. That is okay. Reinstall only the required ESP32-compatible libraries afterward.

10. Verify / Compile

Open FreeWiFi.ino and press Verify (✓). A successful compile means the program and its dependencies are ready to upload. It does not upload video.mp4.

11. Two Different Uploads

```
FreeWiFi.ino  
↓  
ESP32 program flash  
  
data/video.mp4  
↓  
ESP32 LittleFS filesystem
```

Both uploads are required: normal Upload for the .ino program and Upload LittleFS for the video.

12. LittleFS Upload Tool

Use the Arduino LittleFS Upload extension for Arduino IDE 2.x.

Official repository: <https://github.com/earlephilhower/arduino-littlefs-upload>

13. LittleFS Plugin Folder Issue

During setup, the .vsix file was first placed in the wrong folder, plugin-storage. The correct location used was:

```
C:\Users\<YOUR_USERNAME>\.arduinoIDE\plugins\
```

If plugins does not exist, create it and put the .vsix inside.

```
.arduinoIDE/  
  plugin-storage/  
  plugins/  
    arduino-littlefs-upload-xxxxx.vsix  
    ...
```

NOTE: Do not confuse plugins with plugin-storage.

14. Upload LittleFS

Select ESP32 Dev Module and the correct COM port. Press Ctrl + Shift + P, search **Upload LittleFS**, select the command, and wait for successful completion. In the original build this step completed successfully, meaning the video was transferred to LittleFS.

15. Upload the Normal Sketch

After LittleFS succeeds, click the normal Upload (→) button. If uploading remains stuck at Connecting, press and hold the ESP32 BOOT button until uploading begins, then release it.

16. Serial Monitor

Open Tools → Serial Monitor and set 115200 baud. The sketch starts Serial at 115200.

```
LittleFS OK  
Video found!  
Video size: XXXXX bytes  
  
WiFi Started!  
SSID: Free_5G_WiFi_Click_Me  
IP Address: 192.168.4.1  
  
Web server started!
```

17. Wi-Fi Configuration

Setting	Value
SSID	Free_5G_WiFi_Click_Me
Mode	WIFI_AP
ESP32 IP	192.168.4.1
Gateway	192.168.4.1
Subnet	255.255.255.0
HTTP	80
DNS	53
Serial	115200
Video	/video.mp4
Debug	/video-info

18. Captive Portal

The ESP32 starts a DNS server on port 53 and redirects DNS requests to 192.168.4.1. The web server listens on port 80. The sketch contains common captive-portal routes for Android, Apple and Windows.

```
/generate_204  
/hotspot-detect.html
```

```
/ncsi.txt  
/connecttest.txt
```

These routes redirect to the main page. Some phones may not automatically open the portal; manually open <http://192.168.4.1>.

19. Webpage and Video

The webpage is embedded directly in FreeWiFi.ino. There is no separate index.html in this project.

```
<video controls playsinline preload="auto">  
<source src="/video.mp4" type="video/mp4">  
</video>
```

The page also displays the message '■ Got You!'. The browser requests /video.mp4, and the ESP32 serves it from LittleFS as video/mp4.

20. Phone Testing

Connect the phone to **Free_5G_WiFi_Click_Me**. A message such as 'Connected without internet' is normal. If the captive portal does not appear automatically, open:

```
http://192.168.4.1
```

21. Debugging URLs

Main webpage:

```
http://192.168.4.1/
```

Video:

```
http://192.168.4.1/video.mp4
```

Video information:

```
http://192.168.4.1/video-info
```

22. Troubleshooting

Wi-Fi does not appear

- Confirm ESP32 power and USB connection.
- Confirm the sketch uploaded successfully.
- Open Serial Monitor at 115200.
- Look for WiFi Started! and the expected SSID.

LittleFS FAILED!

Upload LittleFS again and confirm data/video.mp4 exists.

video.mp4 NOT FOUND!

Confirm the filename is exactly video.mp4, it is inside data, and run Upload LittleFS again.

Webpage opens but video does not play

- Open /video-info.
- Check that the video exists and has a non-zero size.
- Check Serial Monitor for Video requested and Sending video.
- Try a smaller/different MP4 if needed.

Captive portal does not open automatically

Manually open <http://192.168.4.1>.

Upload stuck at Connecting

Press and hold BOOT until uploading starts, then release it.

23. Changing the Wi-Fi Name

Find the SSID string and replace it, then upload the sketch again.

```
const char* ssid = "Free_5G_WiFi_Click_Me";
```

24. Changing the Video

Replace data/video.mp4 with your own MP4 while keeping the filename video.mp4. Then run Upload LittleFS again. No sketch change is required if the path remains /video.mp4.

25. Storage Considerations

The MP4 must fit in the available LittleFS partition. The ESP32 also needs space for the application and other data. If a large video fails to upload or behaves badly, try a smaller or more compressed MP4.

26. Lessons Learned

- ESP32 and ESP8266 are different platforms; use AsyncTCP for ESP32.
- Library compatibility matters, not only the library name.
- The .ino program and LittleFS files are uploaded separately.
- The data folder must be beside the .ino file.
- The LittleFS .vsix belongs in .arduinoIDE/plugins for this setup, not plugin-storage.
- 'No internet' on the phone is expected for this local AP.
- Serial Monitor at 115200 is the first debugging tool.
- 192.168.4.1 and /video-info are useful manual tests.

27. Complete Rebuild Checklist

1. Install Arduino IDE.
2. Install ESP32 by Espressif Systems.
3. Select ESP32 Dev Module.
4. Install ESPAsyncWebServer by ESP32Async.
5. Install AsyncTCP by ESP32Async.
6. Do NOT install ESPAsyncTCP.
7. Extract the project.
8. Open FreeWiFi.ino.
9. Confirm data/video.mp4 exists.
10. Install Arduino LittleFS Upload.
11. Put the .vsix in .arduinoIDE/plugins.
12. Restart Arduino IDE.
13. Select the correct COM port.
14. Verify the sketch.

15. Upload LittleFS.
16. Wait for successful completion.
17. Upload FreeWiFi.ino normally.
18. Open Serial Monitor at 115200.
19. Confirm LittleFS OK.
20. Confirm Video found!
21. Confirm WiFi Started!
22. Connect phone to Free_5G_WiFi_Click_Me.
23. Open 192.168.4.1 if needed.
24. Test video.
25. Test /video-info.

28. Quick Start

1. Install ESP32 board support.
2. Install ESPAsyncWebServer by ESP32Async.
3. Install AsyncTCP by ESP32Async.
4. Open FreeWiFi.ino.
5. Confirm data/video.mp4.
6. Upload LittleFS.
7. Upload FreeWiFi.ino.
8. Serial Monitor → 115200.
9. Connect to Free_5G_WiFi_Click_Me.
10. Open <http://192.168.4.1>.
11. Play the video.

29. Useful Resources

- Arduino IDE: <https://www.arduino.cc/en/software/>
- ESPAsyncWebServer: <https://github.com/ESP32Async/ESPAsyncWebServer>
- AsyncTCP: <https://github.com/ESP32Async/AsyncTCP>
- Arduino LittleFS Upload: <https://github.com/earlephilhower/arduino-littlefs-upload>
- ESP32 Arduino Core: <https://github.com/espressif/arduino-esp32>

30. Responsible Use

Use the project only on devices and networks where you have permission. Do not use captive portals to impersonate trusted networks, collect credentials, intercept private traffic, or interfere with other people's devices or networks.

31. Final Architecture



```
██████████████████████████████████████████  
█  
▼  
██████████████████████████████████████████  
█ Async Web Server █  
██████████████████████████████████████████  
█  
▼  
██████████████████████████████████████████  
█ LittleFS █  
█ video.mp4 █  
██████████████████████████████████████████
```

NOTE: The most important future reminder: perform both uploads — Upload LittleFS for data/video.mp4, then normal Upload for FreeWiFi.ino. If anything fails, use Serial Monitor at 115200 and test /video-info.